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Prescription Cascades Associated with Acetylcholinesterase Inhibitors Use: A High-Throughput Sequence Symmetry Analysis

Running Title: Prescription Cascades for AChEIs

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Key Points (300 words)

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- Prescription cascades occur where a new drug is prescribed to treat adverse effects caused by another drug; Acetylcholinesterase inhibitors (AChEI) may trigger such cascades due to their known adverse drug effects.
- This study identified potential prescription cascades following initiation of AChEI using high-throughput sequence symmetry analysis.
- Identified signals aligned with gastrointestinal, neuropsychiatric, dermatological, and respiratory adverse effects, supporting the biological plausibility of these prescription cascades.

Why does this paper matter?

This study provides real world evidence of potential prescription cascades associated with AChEI use, highlighting how adverse drug effects may drive prescribing and polypharmacy in dementia patients. By applying high-throughput signal detection, this work supports pharmacovigilance and safer prescribing practices for older adults.

ABSTRACT

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Background: Acetylcholinesterase inhibitors (AChEIs), prescribed for symptomatic dementia, are associated with adverse drug effects which may prompt new drug prescriptions, known as prescription cascades. We aimed to identify potential AChEIs-induced prescription cascades using high-throughput sequence symmetry analysis using real world data.

Methods: Patients aged 18+ with 365 days of prior observation initiating AChEIs (donepezil, rivastigmine and galantamine) were identified from Clinical Practice Research Datalink GOLD between 2002 to 2022. We screened 510 drug classes and 1,213 individual ingredients within ±180 days of AChEIs (365 days as sensitivity). Crude and adjusted sequence ratios (ASR) were calculated with 99% confidence intervals with positive signals undergoing review for clinical plausibility.

Results: We identified 66,155 initiators of AChEIs (median age 81 years [IQR 76 - 85]); 62.8% female). Of the ATC classes and individual ingredients evaluated, 51 and 46 signals were positive with 28 (55%) and 22 (48%) classified as potential prescription cascades after review, respectively. Prescriptions of new drugs acting on the gastrointestinal system showed positive signals such as antipropulsives (ASR 1.50 [1.28 - 1.75]), loperamide (ASR 1.52 [1.30 - 1.77]) and cyclizine (ASR 2.10 [1.72 - 2.59]). Positive signals were found for drugs acting on the nervous system such as benzodiazepine derivatives (ASR 1.83 [1.55 - 2.16]) as well as those acting on the respiratory system such as corticosteroids (ASR 1.66 [1.33 - 2.08]), glucocorticoids (ASR 1.54 [1.30 - 1.83]) and ingredients such as beclomethasone (ASR 1.48 [1.21 - 1.82]). Most positive signals remained in sensitivity analysis, with additional signals mostly related to drugs acting on the nervous system.

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Conclusions: Identified signals suggest potential prescription cascades related to AChEIs use corresponding to gastrointestinal, neuropsychiatric, dermatological and respiratory adverse drug events. While these findings require further validation, this study demonstrates the utility of high-throughput signal detection for identifying potential prescription cascades to support pharmacovigilance in high-risk populations.

Keywords: prescription cascade, sequence symmetry analysis, adverse drug events, acetylcholinesterase inhibitors, pharmacovigilance

INTRODUCTION

Dementia is a growing public health challenge affecting over 57 million people worldwide, with the prevalence projected to grow to 153 million people by 2050(1).

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Acetylcholinesterase inhibitors (AChEIs) (donepezil, rivastigmine, and galantamine) enhance neurotransmission by preventing the breakdown of acetylcholine within synapses. They temporarily stabilize cognitive function and delay progression of dementia related to neurodegenerative diseases such as Alzheimer' disease and Parkinson's disease(2). However, due to their increased stimulation of the parasympathetic nervous system, AChEIs are associated with a range of adverse drug events (ADEs), including gastrointestinal, urinary, neuropsychiatric, and cardiovascular disturbances(3,4). Due to their use as first-line treatments for dementia, millions of patients are potentially introduced to the risk of prescription cascades(5,6).

Prescription cascades can occur when an ADE from a prescribed drug is misinterpreted as a new medical condition, prompting the initiation of additional medications to manage the symptoms(7). This polypharmacy can lead to increased healthcare costs and higher risks of further ADEs, hospitalization and death(8,9), particularly in older adults vulnerable to drug-related harms such as those with dementia(10). Despite the significance of prescription cascades in clinical practice, there is limited research quantifying these events in the context of dementia treatments, specifically among users of AChEIs using real world data.

Sequence Symmetry Analysis (SSA) is a type of self-controlled, case-only screening analysis used for signal detection. It identifies potential associations between two drug events, typically an index drug and a marker drug by comparing the order of these events within individuals, making it useful for detecting prescription cascades and temporal relationships between treatments(11,12). SSA is particularly well-suited for detecting drug-related prescribing patterns in real-world data(13–15), allowing

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identification of known ADEs as well as rarer or unknown ADEs(16). However, to date, no studies have systematically applied this method to investigate prescription cascades involving AChEIs.

The aim of this study is to identify prescription cascades associated with the use of AChEIs using primary care data from the United Kingdom (UK). By employing SSA across both the drug class and ingredients level, we seek to enhance understanding of the downstream prescribing consequences of AChEIs and highlight opportunities for mitigating the risks of prescription cascades including polypharmacy and drug-drug interactions in this high-risk population.

METHODS

Study design, setting, and data sources.

We carried out SSA, a case-only screening analysis, using routinely collected primary care data from the UK. People with a prescription for AChEIs [donepezil, galantamine, and rivastigmine] were identified from the CPRD GOLD database which includes data from GP practices representing ~4.3 % of the UK population(17). CPRD GOLD established in 1987 contains pseudonymized patient-level information on demographics, lifestyle data, clinical diagnoses, prescriptions, and preventive care. To date it covers >21.3 million patients with nearly three million alive and currently registered in contributing GP practices. CPRD GOLD data was mapped to the Observational Medical Outcomes Partnership (OMOP) Common Data Model(18).

Study participants

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Patients were included if they were 18 years or older with a prescription for the index drugs (AChEIs). From these patients, only those with a specific marker of interest (Anatomical Therapeutic Chemical (ATC) class or ingredient) within the last 180 days before or after the index initiation between the study period (January 2003 to January 2023). Only those with 365 days of prior history based on the first prescription of the index and markers were included with a 365-washout applied. Patients were excluded if the date of prescription for the index and marker were on the same date.

Marker definitions

Marker drugs were identified at the drug class level based on the ATC 4th level for the class level analysis (n = 909), and at the ingredient level for the drug level analysis (n = 31,556). We did not include ATC drug classes related to vaccines, vitamins, herbal medicines, food, beverages, supplements, tobacco, and lab tests. For individual drug ingredients, we included both pharmacologically active agents and non-pharmacological substances that may be prescribed (e.g., water, sodium chloride). All ingredients were retained as they may represent markers of clinical events, supportive care, or formulation components.

Drug classes and ingredients with less than 500 patient records were excluded. We also excluded the ATC classes and ingredients representing other anti-dementia drugs. This resulted in 510 drug classes and 1,213 drug ingredients initially available for analysis. A summary of numbers of excluded classes and drug ingredients can be found in the supplement (**Table S1-S2**).

Statistical methods

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We used the R package "CohortSymmetry"(19) to perform SSA to test each ATC class and ingredient in a high throughput manner. For each AChEIs (index)-marker pair, we determined the crude sequence ratio (CSR) as the number of patients who initiated the marker class/ingredient after AChEIs initiation divided by the number of patients who initiated the marker before the AChEIs. Excess initiation of a marker after the AChEIs, relative to before the AChEI, results in a CSR > 1 and may indicate the presence of a prescription cascade. We required the marker initiation to occur within 180 days prior or following AChEIs initiation for the primary analysis with a sensitivity analysis using a longer initiation window of ±365 days.

To account for changes in prescribing trends of index and marker over time, the null sequence ratio (NSR) was calculated for each index-marker pair. Briefly, NSR is the expected sequence ratio in the absence of any causal relationship between the marker and index drug, based on population-level prescribing trends(12,20). The adjusted sequence ratio (ASR) was calculated by dividing the CSR by the NSR for each pair to adjust for background prescribing trends. All ASRs with a lower 99% confidence interval > 1 were considered a positive prescription cascade signal.

To evaluate the performance of SSA before the main study, we first tested positive and negative control drug pairs: amiodarone followed by levothyroxine (positive control) and amiodarone followed by allopurinol (negative control). Additionally, we included a study specific positive control, AChEIs followed by memantine, to reflect a clinically expected treatment sequence in dementia care(21).

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Baseline characteristics of patients including age, sex, specific AChEIs medication, and a predefined list of comorbidities and medications was characterized 180 to 1 day prior to AChEIs initiation(22,23).

All positive signals in the primary analysis were manually reviewed to differentiate potential prescription cascades from false positive signals. False positive signals could be due to various factors such as detection biases, changes in prescription trends, disease progression and reverse causation(14). Manual review of positive signals first involved assessment of the CSRs, ASRs, and NSRs of each marker as well as temporal prescribing patterns over the study period. Secondly, literature searches were used to assess evidence for potential prescription cascades and their underlying mechanisms to support signal classification.

All analytical code for this study is publicly available on GitHub (https://github.com/oxford-pharmacoepi/DementiaPSSA). Analyses were conducted using R version 4.2.3.

RESULTS

We identified 66,155 initiators of AChEIs who were eligible for inclusion. The baseline characteristics of the cohort are summarized in **Table 1**. The majority of AChEIs initiators were female (62.76%) with a median age of 81 years [IQR 76 - 85] (highest proportion of initiators aged 80 to 89 (48.4%)). Donepezil was the most prescribed AChEI (73.5%), followed by rivastigmine (17.9%) and galantamine (15.7%).

Within 180 days preceding AChEI initiation, 65.4% of patients had a dementia diagnosis. The most frequent comorbidities were hypertension (28.1%), renal impairment (19.8%),

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and urinary tract infections (16.5%). Commonly used medications included lipid-modifying agents (41.5%), drugs for acid-related disorders (33.2%), and antidepressants (33.1%).

All study results can be found in a user-friendly interactive web application (https://dpa-pde-oxford.shinyapps.io/dementia_pssa_study/).

Table 1: Baseline characteristics of acetylcholinesterase inhibitor initiators

	CPRD GOLD
Number subjects (N)	66,155
Age	
Median [Q25 - Q75]	81 [76 - 85]
Range	19 to 103
Age group (N %)	
18 to 49	162 (0.24%)
50 to 59	1,029 (1.56%)
60 to 69	5,014 (7.58%)
70 to 79	22,083 (33.4%)
80 to 89	31,987 (48.4%)
90+	5,880 (8.89%)
Sex (N %)	
Female	41,522 (62.8%)
Male	24,633 (37.2%)
Acetylcholinesterase inhibitors (N %)*	
Donepezil	48,625 (73.5%)
Rivastigmine	11,811 (17.9%)
Galantamine	10,358 (15.7%)
Conditions 180 to 1 days prior to index date (N %)	
Dementia	43,240 (65.4%)
Hypertension	18,572 (28.1%)
Renal impairment	13,080 (19.8%)
UTI	10,940 (16.5%)
Anxiety	10,279 (15.5%)
Depressive disorder	10,245 (15.5%)

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Ischemic heart disease	6,998 (10.6%)
Hyperlipidemia	6,882 (10.4%)
Cerebrovascular disease	6,724 (10.2%)
Type 2 Diabetes	6,625 (10.0%)
Osteoporosis	5,899 (8.92%)
Urinary incontinence	5,239 (7.92%)
Atrial fibrillation	4,960 (7.50%)
Asthma	4,455 (6.73%)
Venous thromboembolism	3,304 (4.99%)
COPD	3,290 (4.97%)
Stroke	2,572 (3.89%)
Myocardial infarction	1,910 (2.89%)
GERD	1,741 (2.63%)
Pneumonia	1,731 (2.62%)
Medications 180 to 1 days prior to index date (N %)	
Lipid modifying agents	27,470 (41.5%)
Drugs for acid related disorders	21,983 (33.2%)
Antidepressants	21,864 (33.1%)
Antibacterials systemic	19,700 (29.8%)
Agents for renin ang system	19,360 (29.3%)
Diuretics	17,305 (26.2%)
Antianemic preparations	14,474 (21.9%)
Psycholeptics	14,450 (21.8%)
Opioids	14,423 (21.8%)
Calcium channel blockers	14,132 (21.4%)
Beta blocking agents	12,849 (19.4%)
Antithrombotics	9,373 (14.2%)
Drugs for obstructive airway disorders	8,972 (13.6%)
Anti-inflammatory drugs and antirheumatics	8,538 (12.9%)
Drugs used in diabetes	6,453 (9.75%)
Antiepileptics	4,232 (6.40%)
Immunosuppressants	631 (0.95%)
Antiemetics	163 (0.25%)

* groups are not mutually exclusive

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COPD; Chronic obstructive pulmonary disease, GERD; gastroesophageal reflux disease, UTI; urinary tract infection.

The positive and negative controls were in line with expectations: the positive control amiodarone-levothyroxine drug pair showing a positive ASR of 5.33 [99% CI 4.52 - 6.33], while the negative control amiodarone-allopurinol pair showed a null signal (ASR 0.97 [0.76 - 1.24]). The study-specific positive control AChEIs-memantine pair also showed a positive ASR of 7.18 [5.92 - 8.79].

Among marker ATC drug classes analyzed, there were 51 significant signals identified in the 180-day primary analyses distributed across multiple ATC categories (**Figure 1**). Notable signals within the ATC alimentary tract and metabolism category included A04AD (other antiemetics) with an ASR of 3.64 (99% CI: 2.43 - 5.59), and A03FA (propulsives) with an ASR of 1.58 (1.33–1.87). In the nervous system ATC category, N07BC (drugs used in opioid dependence) had an ASR of 9.86 (5.06 - 21.6) where as N05CM (other hypnotics and sedatives) had an ASR of 2.77 (1.97 - 3.95). Additional signals were observed for N02AB (oripavine derivatives 1.35 [1.08 - 1.70]) and N02AB (phenylpiperidine derivatives 2.67 [1.82–3.99]). Several subgroups in the dermatological ATC class, notably corticosteroids such as D07AB (corticosteroids moderately potent group ii) and D07AC (corticosteroids potent group iii), showed ASRs of 1.20 (1.05 - 1.37) and 1.16 (1.02 - 1.33) respectively. Other signals were observed for systemic anti-infectives such as J01AA (tetracyclines) and J01CA (penicillins with extended spectrum) displayed positive signals. ATC classes from the respiratory system also showed positive signals including R01AD (corticosteroids for nasal use) and R03BA

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(glucocorticoids for obstructive airway diseases) had significant signals with ASRs of 1.66 (1.33 - 2.08) and 1.54 (1.30 - 1.83).

Negative signals (ASR <1) were also observed for some ATC classes (n = 16) including drugs for diabetes (A10BA, biguanides; A10BB, sulfonylureas), cardiovascular drugs (thiazides, ACE inhibitors, dihydropyridine derivatives, HMG-CoA reductase inhibitors), and several other classes such as antithrombotic agents, antianemic preparations, thyroid therapies, salicylic acid derivatives, certain anti-psychotics (diazepines/oxazepines/thiazepines/oxepines), and antidepressants (e.g. selective serotonin reuptake inhibitors) (**Table S3**).

Of the 51 positive signals, 42 (82%) remained positive when the index marker initiation window increased to 365 days, with an additional 17 positive signals not in the primary analysis (**Figure S1**). These additional positive signals included ATC classes for antiemetics (A04AA 1.67 [1.01 - 2.81]) and drugs for constipation (e.g. A06AA 1.15 [1.04 - 1.27]). Several nervous system drugs including antiepileptics (N03AG 1.52 [1.14 - 2.01]), anticholinergic agents (N04AA 1.65 [1.07 - 2.57]), antipsychotics (N05AD 1.54 [1.29 - 1.84], N05AL 1.31 [1.06 - 1.63]), anxiolytics (N05BA) and sedatives (N05CF, N05CH) were also seen as well as iron preparations (B03AB, B03AE), antibacterials (J01DC, J01FA) and nasal preparations (R01AX) (**Table S4**).

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ATC 4th Level	Index N (%)	Marker N (%)	ASR (99% CI)	CSR (99% CI)	NSR
A01AB antiinfectives and antiseptics for local oral treatment	210 (56.00%)	165 (44.00%)	1.36 [1.04 - 1.78]	1.27 [0.97 - 1.67]	0.935
A01AC corticosteroids for local oral treatment	45 (67.20%)	22 (32.80%)	2.38 [1.24 - 4.76]	2.05 [1.07 - 4.10]	0.860
A03AX other drugs for functional gastrointestinal disorders	113 (62.80%)	67 (37.20%)	1.81 [1.22 - 2.70]	1.69 [1.14 - 2.53]	0.934
A03FA propulsives	558 (58.90%)	389 (41.10%)	1.58 [1.33 - 1.87]	1.43 [1.21 - 1.70]	0.911
A04AD other antiemetics	178 (78.40%)	49 (21.60%)	3.64 [2.43 - 5.59]	3.63 [2.43 - 5.59]	0.998
A06AG enemas	289 (56.10%)	226 (43.90%)	1.33 [1.06 - 1.67]	1.28 [1.02 - 1.61]	0.963
A07AA antibiotics	330 (60.30%)	217 (39.70%)	1.59 [1.27 - 2.00]	1.52 [1.22 - 1.91]	0.954
A07DA antipropulsives	667 (58.10%)	481 (41.90%)	1.50 [1.28 - 1.75]	1.39 [1.19 - 1.62]	0.927
B05CB salt solutions	445 (54.10%)	377 (45.90%)	1.33 [1.11 - 1.59]	1.18 [0.99 - 1.41]	0.889
B05XA electrolyte solutions	379 (55.80%)	300 (44.20%)	1.43 [1.17 - 1.75]	1.26 [1.04 - 1.54]	0.883
C05AX other agents for treatment of hemorrhoids and anal fissures for topical use	617 (56.30%)	478 (43.70%)	1.41 [1.20 - 1.64]	1.29 [1.10 - 1.51]	0.919
D01AC imidazole and triazole derivatives	643 (53.90%)	551 (46.10%)	1.24 [1.07 - 1.44]	1.17 [1.01 - 1.36]	0.939
D06AX other antibiotics for topical use	488 (55.20%)	396 (44.80%)	1.32 [1.11 - 1.57]	1.23 [1.04 - 1.47]	0.934
D07AB corticosteroids moderately potent group ii	797 (52.70%)	716 (47.30%)	1.20 [1.05 - 1.37]	1.11 [0.97 - 1.27]	0.931
D07AC corticosteroids potent group iii	764 (52.10%)	702 (47.90%)	1.16 [1.02 - 1.33]	1.09 [0.95 - 1.25]	0.935
D07BA corticosteroids weak combinations with antiseptics	303 (54.40%)	254 (45.60%)	1.25 [1.01 - 1.56]	1.19 [0.96 - 1.49]	0.951
D07CB corticosteroids moderately potent combinations with antibiotics	26 (66.70%)	13 (33.30%)	2.46 [1.05 - 6.16]	2.00 [0.86 - 5.01]	0.813
D07XA corticosteroids weak other combinations	1,014 (51.60%)	952 (48.40%)	1.15 [1.02 - 1.29]	1.07 [0.95 - 1.20]	0.928
D07XB corticosteroids moderately potent other combinations	187 (56.70%)	143 (43.30%)	1.48 [1.11 - 1.98]	1.31 [0.98 - 1.74]	0.883
D09AA medicated dressings with antiinfectives	469 (56.50%)	361 (43.50%)	1.37 [1.15 - 1.64]	1.30 [1.09 - 1.56]	0.947
D09AB zinc bandages	568 (57.30%)	423 (42.70%)	1.44 [1.22 - 1.70]	1.34 [1.14 - 1.59]	0.930
D11AC medicated shampoos	881 (57.10%)	663 (42.90%)	1.39 [1.22 - 1.58]	1.33 [1.16 - 1.52]	0.958
H02AB glucocorticoids	549 (53.20%)	482 (46.80%)	1.19 [1.02 - 1.40]	1.14 [0.97 - 1.34]	0.954
J01AA tetracyclines	677 (53.10%)	599 (46.90%)	1.17 [1.01 - 1.35]	1.13 [0.98 - 1.31]	0.967
J01CA penicillins with extended spectrum	1,320 (52.00%)	1,218 (48.00%)	1.19 [1.07 - 1.32]	1.08 [0.98 - 1.20]	0.910
J01CR combinations of penicillins incl beta lactamase inhibitors	825 (52.20%)	755 (47.80%)	1.21 [1.06 - 1.38]	1.09 [0.96 - 1.24]	0.903
J01DB first generation cephalosporins	710 (52.70%)	637 (47.30%)	1.28 [1.11 - 1.47]	1.11 [0.97 - 1.28]	0.871
J01EA trimethoprim and derivatives	1,877 (50.70%)	1,826 (49.30%)	1.10 [1.01 - 1.20]	1.03 [0.94 - 1.12]	0.934
J05AH neuraminidase inhibitors	26 (70.30%)	11 (29.70%)	2.55 [1.05 - 6.81]	2.36 [0.97 - 6.32]	0.927
M01AC oxicams	80 (58.80%)	56 (41.20%)	1.64 [1.05 - 2.58]	1.43 [0.92 - 2.25]	0.873
N02AA natural opium alkaloids	1,102 (51.50%)	1,039 (48.50%)	1.15 [1.03 - 1.28]	1.06 [0.95 - 1.19]	0.923
N02AB phenylpiperidine derivatives	151 (71.20%)	61 (28.80%)	2.67 [1.82 - 3.99]	2.48 [1.69 - 3.70]	0.927
N02AE oripavine derivatives	305 (56.90%)	231 (43.10%)	1.35 [1.08 - 1.70]	1.32 [1.06 - 1.65]	0.976
N02BE anilides	1,689 (50.90%)	1,627 (49.10%)	1.14 [1.05 - 1.25]	1.04 [0.95 - 1.14]	0.908
N05AA phenothiazines with aliphatic side chain	299 (64.20%)	167 (35.80%)	1.82 [1.42 - 2.34]	1.79 [1.40 - 2.30]	0.984
N05AX other antipsychotics	642 (52.60%)	578 (47.40%)	1.17 [1.01 - 1.35]	1.11 [0.96 - 1.29]	0.952
N05CD benzodiazepine derivatives	630 (62.30%)	382 (37.70%)	1.83 [1.55 - 2.16]	1.65 [1.40 - 1.95]	0.902
N05CM other hypnotics and sedatives	205 (73.20%)	75 (26.80%)	2.77 [1.97 - 3.95]	2.73 [1.95 - 3.90]	0.988
N07BC drugs used in opioid dependence	138 (90.80%)	14 (9.20%)	10.34 [5.31 - 22.68]	9.86 [5.06 - 21.63]	0.954
P01AB nitroimidazole derivatives	266 (57.50%)	197 (42.50%)	1.44 [1.14 - 1.84]	1.35 [1.06 - 1.72]	0.935
P01BC methanolquinolines	310 (60.10%)	206 (39.90%)	1.69 [1.34 - 2.14]	1.50 [1.20 - 1.90]	0.889
P03AX other ectoparasiticides incl scabicides	39 (67.20%)	19 (32.80%)	2.40 [1.19 - 5.09]	2.05 [1.02 - 4.34]	0.854
R01AD corticosteroids	348 (61.40%)	219 (38.60%)	1.66 [1.33 - 2.08]	1.59 [1.27 - 1.99]	0.957
R03BA glucocorticoids	555 (59.10%)	384 (40.90%)	1.54 [1.30 - 1.83]	1.45 [1.22 - 1.72]	0.937
R03BB anticholinergics	177 (55.80%)	140 (44.20%)	1.36 [1.02 - 1.82]	1.26 [0.95 - 1.69]	0.931
R06AE piperazine derivatives	780 (61.30%)	493 (38.70%)	1.66 [1.43 - 1.93]	1.58 [1.37 - 1.84]	0.952
R06AX other antihistamines for systemic use	264 (54.90%)	217 (45.10%)	1.30 [1.03 - 1.65]	1.22 [0.96 - 1.54]	0.933
S01AA antibiotics	738 (52.30%)	674 (47.70%)	1.20 [1.05 - 1.38]	1.09 [0.95 - 1.26]	0.910
S01KA viscoelastic substances	305 (55.70%)	243 (44.30%)	1.27 [1.02 - 1.58]	1.26 [1.01 - 1.57]	0.990
S01XA other ophthalmologicals	408 (52.40%)	370 (47.60%)	1.22 [1.02 - 1.47]	1.10 [0.92 - 1.33]	0.902
S03AA antiinfectives	767 (52.90%)	683 (47.10%)	1.23 [1.08 - 1.41]	1.12 [0.98 - 1.29]	0.911

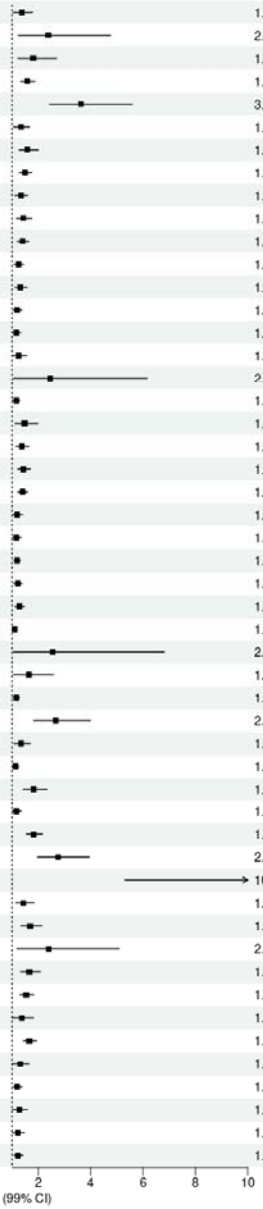

Figure 1: Forest Plot of positive signals from high-throughput sequence symmetry analysis of Acetylcholinesterase inhibitor versus ATC Drug Classes (4th level) with a 180-day initiation window (ASR: adjusted sequence ratio, CSR: crude sequence ratio, NSR: null sequence ratio)

Among marker ingredients analyzed, there were 46 positive signals identified in the 180-day primary analyses (**Figure 2**). Positive signals spanned various ingredients

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including analgesics, gastrointestinal agents, sedatives, corticosteroids, and antipsychotics. Positive signals included opioids such as morphine (ASR 2.54 [2.03 - 3.20]), fentanyl (2.75 [1.87 - 4.13]), and buprenorphine (1.35 [1.08 - 1.70]). Hypnotics and sedatives such as midazolam (11.20 [6.89 - 19.39]) and methotrimeprazine (18.18 [8.25 - 48.64]) also were positive. Other ingredients included corticosteroids (beclomethasone 1.48 [1.21 - 1.82], fluticasone 1.51 [1.14 - 1.99]), antibiotics (trimethoprim 1.10 [1.01- 1.20], metronidazole 1.42 [1.15 -1.76], amoxicillin 1.20 [1.08 - 1.33]), and gastrointestinal agents such as domperidone (1.54 [1.24 - 1.91]) and loperamide (1.52 [1.30 - 1.77]). Non-pharmaceutical agents such as water (8.47 [5.07 - 15.11]) and sodium chloride (1.27 [1.08 - 1.51]) also showed positive signals.

Negative signals (n = 31) were again observed across several therapeutic classes including cardiovascular and anticoagulant agents (e.g., aspirin, amlodipine, apixaban), lipid-lowering therapies (e.g., atorvastatin, simvastatin), antidiabetic drugs (e.g., metformin, gliclazide), thyroid and hematological ingredients (e.g., levothyroxine, vitamin B12), and psychotropic medications (e.g., sertraline, quetiapine) (**Table S5**).

Increasing the initiation window to 365 days, 38 (83%) remained positive with 19 additional positive signals compared to the primary analysis (**Figure S2**). These additional positive signals included psychotropic and neurological drugs such as amisulpride (ASR 1.30 [1.04–1.64]), trazodone (1.44 [1.23–1.69]), valproate (1.51 [1.14–2.01]), haloperidol (1.53 [1.28–1.83]), and lorazepam (1.58 [1.41–1.77]). Other ingredients included antiemetics (ondansetron), laxatives (sennoside B, docusate), respiratory agents (ipratropium), and analgesics (oxycodone) (**Table S6**).

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Drug Ingredient	Index N (%)	Marker N (%)	ASR (99% CI)	CSR (99% CI)	NSR
Methotrimeprazine	173 (95.10%)	9 (4.90%)	18.18 [8.25 - 48.64]	19.22 [8.72 - 51.44]	1.058
Heroin	130 (92.90%)	10 (7.10%)	13.27 [6.17 - 33.86]	13.00 [6.04 - 33.16]	0.979
Midazolam	319 (92.20%)	27 (7.80%)	11.20 [6.89 - 19.39]	11.81 [7.27 - 20.47]	1.055
Water	216 (89.60%)	25 (10.40%)	8.47 [5.07 - 15.11]	8.64 [5.17 - 15.41]	1.020
Glycopyrronium	90 (82.60%)	19 (17.40%)	4.39 [2.37 - 8.77]	4.74 [2.56 - 9.46]	1.079
Scopolamine	178 (78.40%)	49 (21.60%)	3.63 [2.43 - 5.59]	3.63 [2.43 - 5.59]	0.999
Dicyclomine	21 (75.00%)	7 (25.00%)	3.55 [1.24 - 12.10]	3.00 [1.05 - 10.21]	0.844
Fentanyl	152 (72.00%)	59 (28.00%)	2.75 [1.87 - 4.13]	2.58 [1.75 - 3.86]	0.936
Povidone Iodine	36 (69.20%)	16 (30.80%)	2.61 [1.23 - 5.86]	2.25 [1.07 - 5.06]	0.863
SLSA	54 (67.50%)	26 (32.50%)	2.56 [1.40 - 4.83]	2.08 [1.14 - 3.92]	0.812
Oseltamivir	26 (70.30%)	11 (29.70%)	2.55 [1.05 - 6.81]	2.36 [0.97 - 6.32]	0.927
Morphine	460 (72.00%)	179 (28.00%)	2.54 [2.03 - 3.20]	2.57 [2.05 - 3.24]	1.012
Malathion	32 (66.70%)	16 (33.30%)	2.40 [1.12 - 5.46]	2.00 [0.93 - 4.55]	0.834
Butylscopolamine	293 (69.80%)	127 (30.20%)	2.30 [1.76 - 3.04]	2.31 [1.76 - 3.05]	1.001
Permethrin	96 (66.70%)	48 (33.30%)	2.16 [1.38 - 3.45]	2.00 [1.28 - 3.19]	0.925
Cyclizine	499 (68.40%)	230 (31.60%)	2.10 [1.72 - 2.59]	2.17 [1.77 - 2.67]	1.031
Polyethylene Glycols	54 (64.30%)	30 (35.70%)	2.09 [1.18 - 3.82]	1.80 [1.01 - 3.29]	0.860
Peppermint Oil	99 (65.60%)	52 (34.40%)	2.00 [1.30 - 3.14]	1.90 [1.23 - 2.99]	0.953
Procyclidine	40 (64.50%)	22 (35.50%)	2.00 [1.03 - 4.06]	1.82 [0.93 - 3.68]	0.907
Quinine	309 (60.00%)	206 (40.00%)	1.68 [1.33 - 2.12]	1.50 [1.19 - 1.89]	0.894
Meloxicam	78 (59.50%)	53 (40.50%)	1.66 [1.06 - 2.65]	1.47 [0.93 - 2.34]	0.885
Triamcinolone	91 (58.70%)	64 (41.30%)	1.61 [1.06 - 2.47]	1.42 [0.94 - 2.18]	0.882
Alginic Acid	73 (54.90%)	60 (45.10%)	1.58 [1.01 - 2.49]	1.22 [0.78 - 1.91]	0.769
Nystatin	325 (60.30%)	214 (39.70%)	1.58 [1.26 - 1.99]	1.52 [1.21 - 1.91]	0.960
Metoclopramide	303 (59.50%)	206 (40.50%)	1.57 [1.25 - 1.99]	1.47 [1.17 - 1.86]	0.935
Cephradine	85 (56.30%)	66 (43.70%)	1.56 [1.03 - 2.39]	1.29 [0.85 - 1.97]	0.825
Domperidone	338 (57.80%)	247 (42.20%)	1.54 [1.24 - 1.91]	1.37 [1.10 - 1.70]	0.889
Loperamide	660 (58.60%)	466 (41.40%)	1.52 [1.30 - 1.77]	1.42 [1.21 - 1.66]	0.934
Fluticasone	211 (59.30%)	145 (40.70%)	1.51 [1.14 - 1.99]	1.46 [1.10 - 1.93]	0.967
Beclomethasone	387 (58.20%)	278 (41.80%)	1.48 [1.21 - 1.82]	1.39 [1.14 - 1.71]	0.938
Psyllium	129 (54.00%)	110 (46.00%)	1.46 [1.05 - 2.05]	1.17 [0.84 - 1.64]	0.801
Metronidazole	338 (57.10%)	254 (42.90%)	1.42 [1.15 - 1.76]	1.33 [1.08 - 1.65]	0.938
Loratadine	187 (57.00%)	141 (43.00%)	1.41 [1.06 - 1.89]	1.33 [1.00 - 1.77]	0.938
Buprenorphine	305 (56.90%)	231 (43.10%)	1.35 [1.08 - 1.70]	1.32 [1.06 - 1.65]	0.975
Mometasone	313 (55.20%)	254 (44.80%)	1.30 [1.05 - 1.62]	1.23 [0.99 - 1.53]	0.949
Fusidate	584 (54.20%)	493 (45.80%)	1.29 [1.10 - 1.51]	1.18 [1.01 - 1.39]	0.922
Cephalexin	640 (52.80%)	571 (47.20%)	1.28 [1.10 - 1.48]	1.12 [0.97 - 1.30]	0.876
Sodium Chloride	501 (53.00%)	444 (47.00%)	1.27 [1.08 - 1.51]	1.13 [0.95 - 1.34]	0.886
Chloramphenicol	694 (53.90%)	593 (46.10%)	1.27 [1.10 - 1.47]	1.17 [1.01 - 1.35]	0.920
Cetirizine	401 (53.80%)	344 (46.20%)	1.23 [1.02 - 1.49]	1.17 [0.96 - 1.41]	0.945
Risperidone	583 (52.70%)	523 (47.30%)	1.21 [1.03 - 1.41]	1.11 [0.95 - 1.30]	0.925
Amoxicillin	1,304 (52.20%)	1,195 (47.80%)	1.20 [1.08 - 1.33]	1.09 [0.98 - 1.21]	0.911
Doxycycline	688 (54.20%)	581 (45.80%)	1.19 [1.03 - 1.38]	1.18 [1.02 - 1.37]	0.993
Acetaminophen	1,852 (51.80%)	1,725 (48.20%)	1.17 [1.07 - 1.27]	1.07 [0.98 - 1.17]	0.921
Lactulose	905 (52.10%)	831 (47.90%)	1.17 [1.04 - 1.33]	1.09 [0.96 - 1.23]	0.928
Trimethoprim	1,877 (50.70%)	1,826 (49.30%)	1.10 [1.01 - 1.20]	1.03 [0.94 - 1.12]	0.934

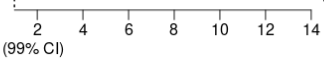

Figure 2: Forest Plot of positive signals from high-throughput sequence symmetry analysis of Acetylcholinesterase inhibitor versus drug ingredients with

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a 180-day initiation window (ASR: adjusted sequence ratio, CSR: crude sequence ratio, NSR: null sequence ratio)

Review of positive signals across ATC classes and ingredients determined 28 (55%) and 22 (48%) signals were potential prescription cascades. The review of the positive signals and their prioritization can be found in the supplement (**Table S7 and S8**). The review of the additional positive signals from the 365-window sensitivity analysis is also provided in the supplement (**Table S9 and S10**).

DISCUSSION

This high-throughput SSA identified several drug classes and ingredients whose initiation following AChEIs were consistent with possible prescription cascades, where side effects or complications from AChEIs prompted additional medication use. Many positive signals were linked to known adverse effects of AChEIs, particularly gastrointestinal, dermatological, neuropsychiatric, and respiratory systems.

AChEIs inhibit acetylcholinesterase, preventing the rapid breakdown of acetylcholine in the peripheral nervous system(24). The accumulation of acetylcholine leads to overstimulation of cholinergic receptions leading to several adverse events including diarrhea, nausea, vomiting, dizziness, muscle cramping, tremors, insomnia, urinary incontinence and seizures with evidence showing commonly reported ADEs are neuropsychiatric, gastrointestinal, and cardiovascular in nature(25). This pharmacological profile provides a plausible biological rationale for several positive signals observed in this work.

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Increasing acetylcholine levels cause stimulation of muscarinic receptors in the nasal passages and airways, leading to increased mucus production, rhinorrhea, bronchial secretions, and bronchoconstriction(26,27); supported by this work and others with signals for medications used to counter these effects such as antihistamines, glucocorticoids and corticosteroids(28). Gastrointestinal related drugs reflect treatments for common cholinergic side effects such as nausea, vomiting, and motility disturbances(29,30). Furthermore, signals related to dermatological drugs and ingredients, including topical corticosteroids suggest management of drug-induced skin reactions(29), however, could be due to consequences of aging and associated care(31). Psychiatric conditions and treatments are frequently associated with dementia(32). Positive signals may reflect both side effects of AChEIs(33) and progression of the underlying disease(34). This interpretation is supported by our sensitivity analysis, which identified additional neurological medications using a wider initiation window. Consistent signals across both windows strengthen confidence in these associations, whereas discrepancies may also indicate differences between acute and delayed effects or prescribing practices related to disease progression.

Although signals obtained in this work align with known side effects some positive signals, such as systemic antibiotics and antiparasitic agents, are more likely attributable to comorbid infections rather than direct prescription cascades related to AChEIs initiation. However, there is conflicting evidence regarding AChEIs and infection risk(35,36). Furthermore, evidence shows AChEIs are associated with urinary incontinence leading to prescription of antimuscarinic medications, with no positive signals observed in our study. This could reflect under-ascertainment of events,

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variability in prescribing, awareness of polypharmacy and anticholinergic as well as individual AChEIs drug effects. A comparative effectiveness study showed higher prevalence of antimuscarinic prescriptions comparing donepezil to rivastigmine(37) whereas another study showed no difference between patients with dementia prescribed AChEIs with those not prescribed them(38). Similarly, a study identified an positive association between beta blockers and AChEI using prescription data from the Netherlands (39), whereas our analysis found a negative signal. This difference is likely due to the shorter initiation window of 180 days used for this work compared to the two years used by the other study. Longer initiation windows can give rise to time varying confounding with careful consideration of appropriate windows considering the disease specific context with sensitivity analysis required. Differences in population characteristics or prescribing patterns of AChEIs may also explain this discrepancy.

A small number of negative signals were observed in this study. While these should be interpreted cautiously, such findings could suggest either protective prescribing patterns, competing risks, or clinical avoidance of certain medications in patients receiving AChEIs, with clinical guidance recommending reviewing and deprescribing medications to prevent inappropriate prescription cascades in older populations(40).

Beyond acting as proxies for adverse effects, prescription cascades are also clinically important. The initiation of additional treatments to manage side effects contributes to polypharmacy, increasing the risk of drug–drug interactions, adverse drug events, and treatment burden(41). Drugs such as benzodiazepines, which may increase sedation and fall risk(42), gastrointestinal agents such as domperidone, which are associated with QT prolongation(43) showed positive signals and illustrate how prescription

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cascades may contribute to polypharmacy and drug-drug interactions following AChEI initiation. In older adults with dementia, such prescription cascades may further increase the risk of adverse events and contribute to cognitive decline(44).

Strengths of this study include the use of a large, comprehensive electronic health records database that enabled real-world pharmacoepidemiologic signal detection across a large population. The use of SSA reduced bias from time-invariant confounders and accounted for certain forms of time-varying confounding via adjustment of prescribing trends. Analyses were conducted at both the drug class (ATC level) and individual ingredient level, allowing identification of patterns that may be obscured at the class level and decreased power at the ingredient level offering complementary insights.

However, our study has some limitations. SSA is a signal detection method used to detect temporal asymmetry; therefore, it cannot be used to establish causality, with further validation of positive signals required using more robust epidemiological methods(45). Some signals may reflect disease progression rather than prescription cascades. For example, increased use of antipsychotics or sedatives could be driven by worsening dementia symptoms, which may occur following initiation of AChEIs(46,47). Prescription records in primary care identify if a medication was prescribed, but they do not confirm whether the patient took the drug or adhered to the treatment regimen. Additionally, as AChEIs are often initiated in specialist care before appearing in primary-care records, early adverse events or prescribing cascades that may occur might be missed leading to misclassification and attenuation of signals. We did not restrict our population to individuals over 65 years of age, based on the assumption that the

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biological processes underlying the development of side effects from AChEIs would not differ by age with most of the population in this study being older any attenuation due to including younger individuals was expected to be minimal. Finally, some positive signals may reflect confounding by indication, reverse causation and observational period imbalance.

CONCLUSION

In summary, this study identified potential prescription cascades associated with initiation of AChEIs. Many associations corresponded with known ADEs, supporting the biological plausibility of these findings, while others may reflect disease progression or indirect prescribing behaviors. These findings demonstrate the value of high-throughput signal detection for identifying potential prescription cascades to help inform pharmacovigilance in high-risk populations. Further studies using complementary epidemiologic designs are warranted to confirm and validate findings.

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Sponsor Role: The sponsors of the study did not have any involvement in the writing of the manuscript or the decision to submit it for publication.

DATA AVAILIABILITY STATEMENT

This study is based in part on data from the Clinical Practice Research Datalink (CPRD) obtained under licence from the UK Medicines and Healthcare products Regulatory Agency. The data is provided by patients and collected by the NHS as part of their care and support. The interpretation and conclusions contained in this study are those of the author/s alone. Patient level data used in this study was obtained through an approved application-the CPRD (application number 23_003334) and is only available following an approval process-safeguard the confidentiality of patient data. Details on how to apply for data access can be found at https://cprd.com/data-access.

Individual patient consent was not required, as CPRD data are de-identified and provided under ethical approval from the UK Health Research Authority (HRA) and the NHS Health and Social Care Research Ethics Committee.

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TABLES

Table 1: Baseline characteristics of acetylcholinesterase inhibitor initiators

	CPRD GOLD
Number subjects (N)	66,155
Age	
Median [Q25 - Q75]	81 [76 - 85]
Range	19 to 103
Age group (N %)	
18 to 49	162 (0.24%)

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50 to 59	1,029 (1.56%)
60 to 69	5,014 (7.58%)
70 to 79	22,083 (33.4%)
80 to 89	31,987 (48.4%)
90+	5,880 (8.89%)
Sex (N %)	
Female	41,522 (62.8%)
Male	24,633 (37.2%)
Acetylcholinesterase inhibitors (N %)	
Donepezil	48,625 (73.5%)
Rivastigmine	11,811 (17.9%)
Galantamine	10,358 (15.67%)
Conditions 180 to 1 days prior to index date (N %)	
Dementia	43,240 (65.4%)
Hypertension	18,572 (28.1%)
Renal impairment	13,080 (19.8%)
UTI	10,940 (16.5%)
Anxiety	10,279 (15.5%)
Depressive disorder	10,245 (15.5%)
Ischemic heart disease	6,998 (10.6%)
Hyperlipidemia	6,882 (10.4%)
Cerebrovascular disease	6,724 (10.2%)
Type 2 Diabetes	6,625 (10.0%)
Osteoporosis	5,899 (8.92%)
Urinary incontinence	5,239 (7.92%)
Atrial fibrillation	4,960 (7.50%)
Asthma	4,455 (6.73%)
Venous thromboembolism	3,304 (4.99%)
COPD	3,290 (4.97%)
Stroke	2,572 (3.89%)
Myocardial infarction	1,910 (2.89%)
GERD	1,741 (2.63%)
Pneumonia	1,731 (2.62%)
Medications 180 to 1 days prior to index date (N %)	
Lipid modifying agents	27,470 (41.5%)
Drugs for acid related disorders	21,983 (33.2%)
Antidepressants	21,864 (33.1%)

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Antibacterials systemic	19,700 (29.8%)
Agents for renin ang system	19,360 (29.3%)
Diuretics	17,305 (26.2%)
Antianemic preparations	14,474 (21.9%)
Psycholeptics	14,450 (21.8%)
Opioids	14,423 (21.8%)
Calcium channel blockers	14,132 (21.4%)
Beta blocking agents	12,849 (19.4%)
Antithrombotics	9,373 (14.2%)
Drugs for obstructive airway disorders	8,972 (13.6%)
Anti-inflammatories and antirheumatics	8,538 (12.9%)
Drugs used in diabetes	6,453 (9.75%)
Antiepileptics	4,232 (6.40%)
Immunosuppressants	631 (0.95%)
Antiemetics	163 (0.25%)

* groups are not mutually exclusive

COPD; Chronic obstructive pulmonary disease, GERD; gastroesophageal reflux disease, UTI; urinary tract infection.

LEGENDS

Figure 1: Forest Plot of positive signals from high-throughput sequence symmetry analysis of Acetylcholinesterase inhibitor versus ATC Drug Classes (4th level) with a 180-day initiation window (ASR: adjusted sequence ratio, CSR: crude sequence ratio, NSR: null sequence ratio)

Figure 2: Forest Plot of positive signals from high-throughput sequence symmetry analysis of Acetylcholinesterase inhibitor versus drug ingredients with

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a 180-day initiation window (ASR: adjusted sequence ratio, CSR: crude sequence ratio, NSR: null sequence ratio)